

Experiment Report: Audio Digitization & Quantization Effects

Date: December 11, 2025

Equipment: ESP32 Microcontroller, INMP441 Microphone (I2S)

1. Experiment Objective

The objective of this experiment is to study and compare the effects of sampling rate and quantization depth on the perceptual quality of recorded audio. By manipulating sampling frequency and bit depth using the ESP32, we aim to develop intuition about how digital audio quality is influenced by temporal resolution and amplitude resolution.

2. Methodology & Tasks

Following the experiment guidelines, the following tasks were executed:

- Capture:** 32,000 audio samples were captured from the INMP441 at a base rate of 16 kHz.
 - Sampling Variation:** The audio was downsampled to create three datasets:
 - 16 kHz:** Baseline (Natural speech)
 - 8 kHz:** Every 2nd sample retained (Telephony quality)
 - 4 kHz:** Every 4th sample retained (Muffled quality)
 - Quantization Variation:** The bit depth was reduced to create three variations:
 - 16-bit:** Baseline (65,536 levels).
 - 8-bit:** Reduced dynamic range (256 levels).
 - 4-bit:** Minimal dynamic range (16 levels).
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3. Results: Data Size Comparison

The following table summarizes the relationship between sampling parameters and the resulting data size for a 2-second clip (32,000 samples at baseline).

Parameter	Configuration	Samples Collected	Resulting File Name Examples
Sampling Rate	16 kHz (Baseline), 8 kHz, 4 kHz	32,000, 16,000, 8,000	rec_16000Hz_16bit.wav, rec_8000Hz_16bit.wav, rec_4000Hz_16bit.wav
Bit Depth	16-bit (Baseline), 8-bit, 4-bit	16-bit (2 bytes), 8-bit (1 byte), 4-bit (0.5 byte)	rec_16000Hz_16bit.wav, rec_16000Hz_8bit.wav, rec_16000Hz_4bit.wav

Data Size Analysis Table

The table below calculates the **Amount of Data in Bytes** for a **2-second recording**.

Formula Used:

$$\text{Data (Bytes)} = \text{Sampling Rate (Hz)} \times \text{Duration (s)} \times \text{Bit Depth}/8$$

Sampling Rate \ Bit Resolution	16-bit (2 Bytes/sample)	8-bit (1 Byte/sample)	4-bit (0.5 Bytes/sample)
16 kHz	(64 KB)	(32 KB)	(16 KB)
8 kHz	(32 KB)	(16 KB)	(8 KB)
4 kHz	(16 KB)	(8 KB)	(4 KB)

The table demonstrates a linear relationship between audio quality parameters and storage requirements:

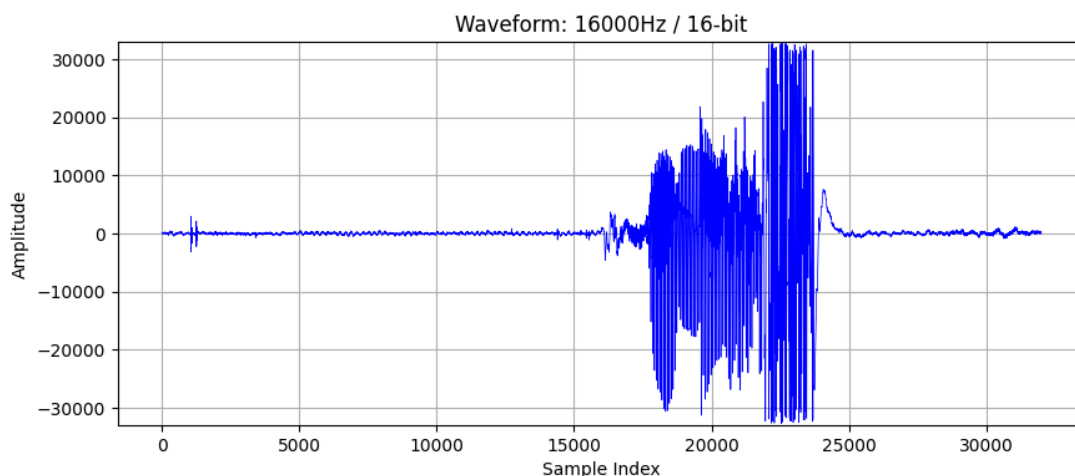
- **Sampling Rate Impact:** Halving the sampling rate (e.g., from 16 kHz to 8 kHz) exactly halves the number of samples collected, and thus halves the file size.
 - **Bit Depth Impact:** Halving the bit depth (e.g., from 16-bit to 8-bit) halves the data required per sample, which also halves the total file size.
 - **Combined Impact:** The most aggressive compression (4 kHz / 4-bit) produces a file size of only **4 KB**, which is **1/16th** the size of the baseline recording (64 KB). However, this extreme reduction comes at the cost of severe quality degradation.
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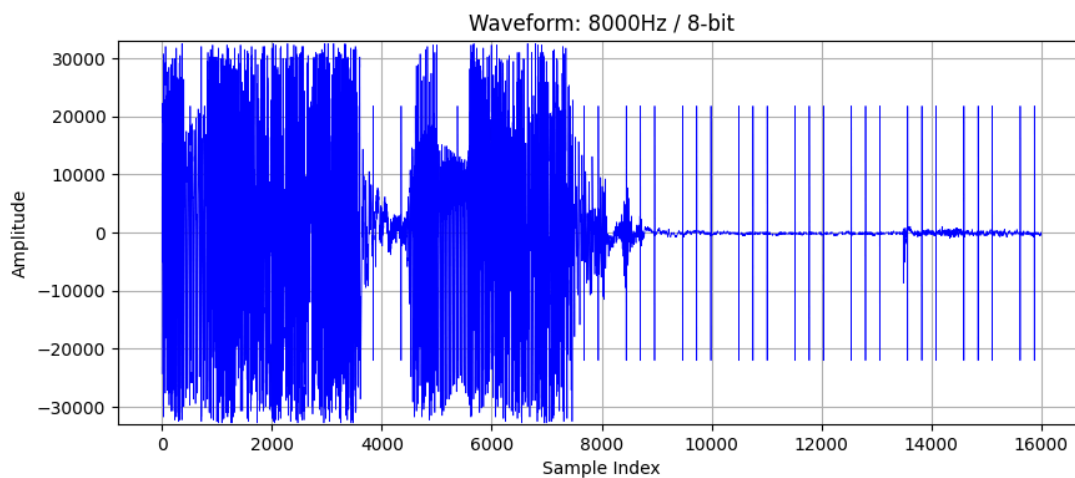
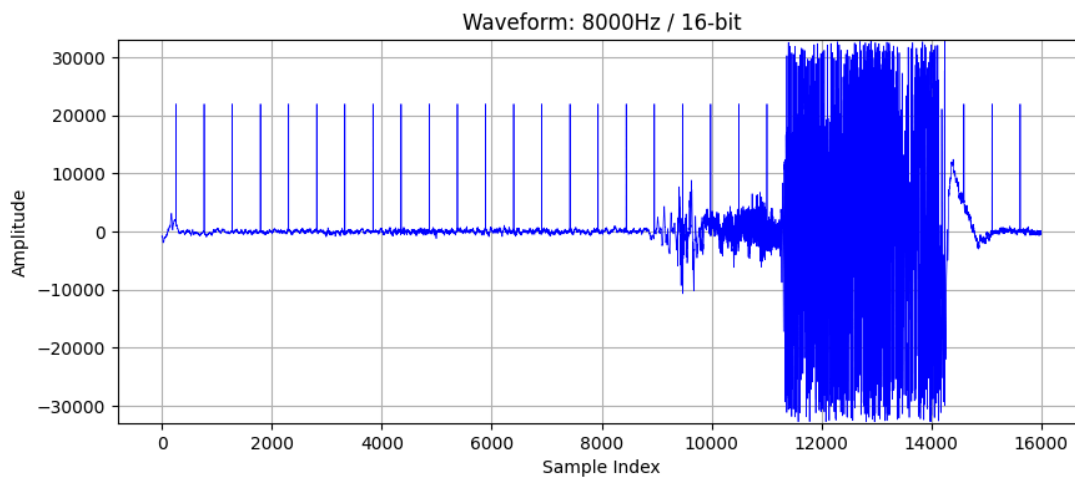
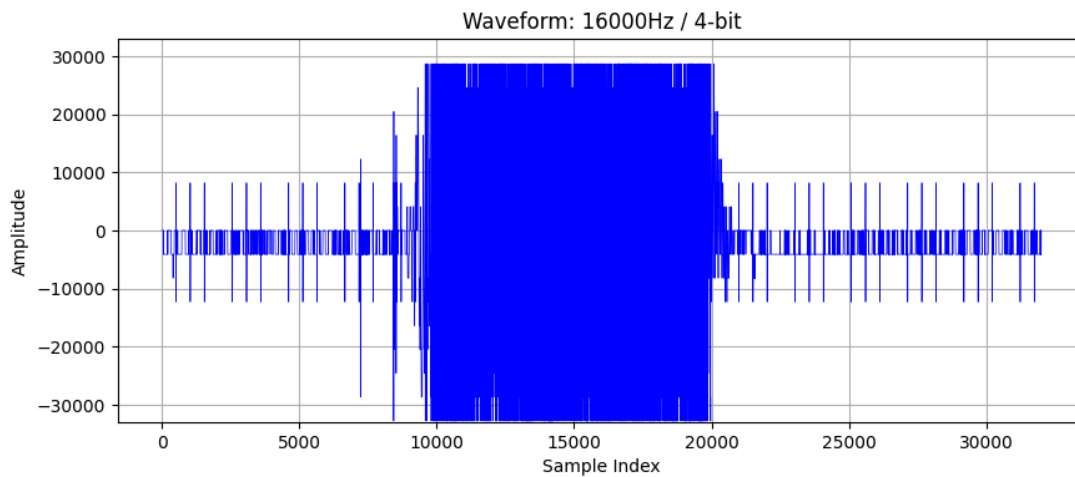
4. Visual Reconstruction

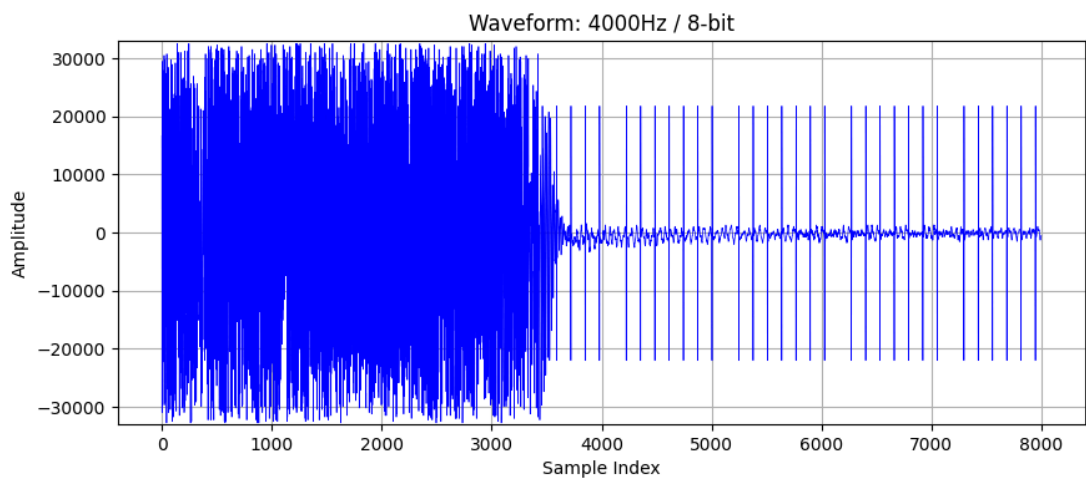
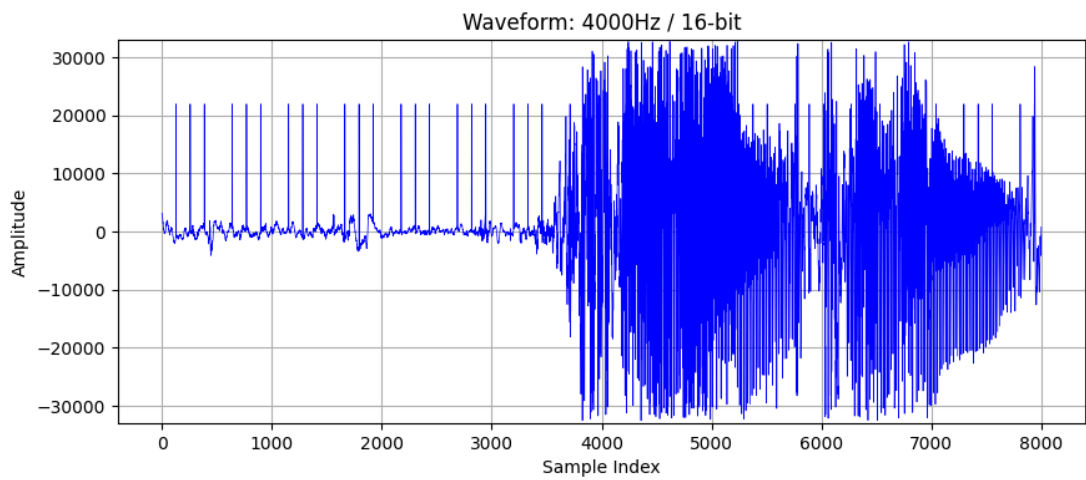
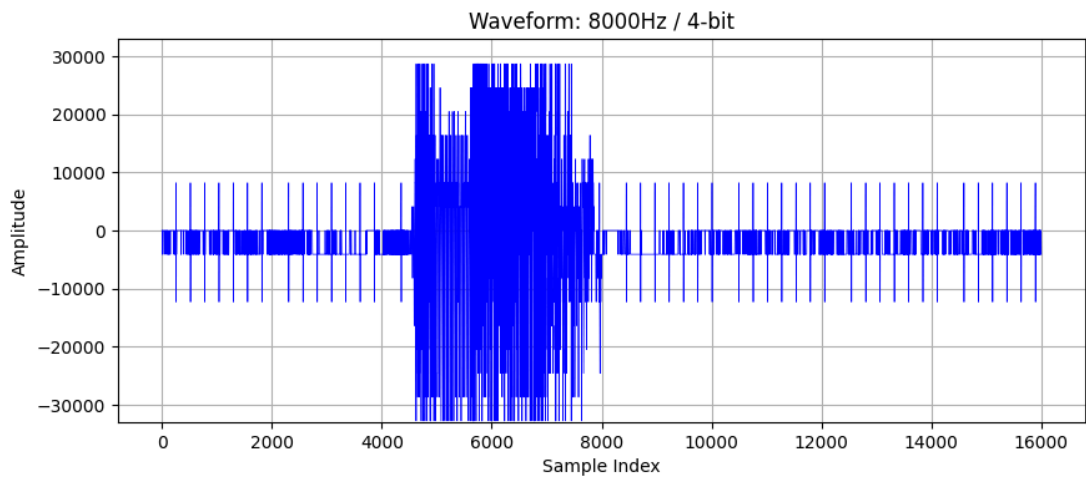
The Python script was used to plot the reconstructed waveforms. The comparison below illustrates the loss of detail (steps) in lower bit-depth signals and the loss of high-frequency sharpness in lower sampling rates.

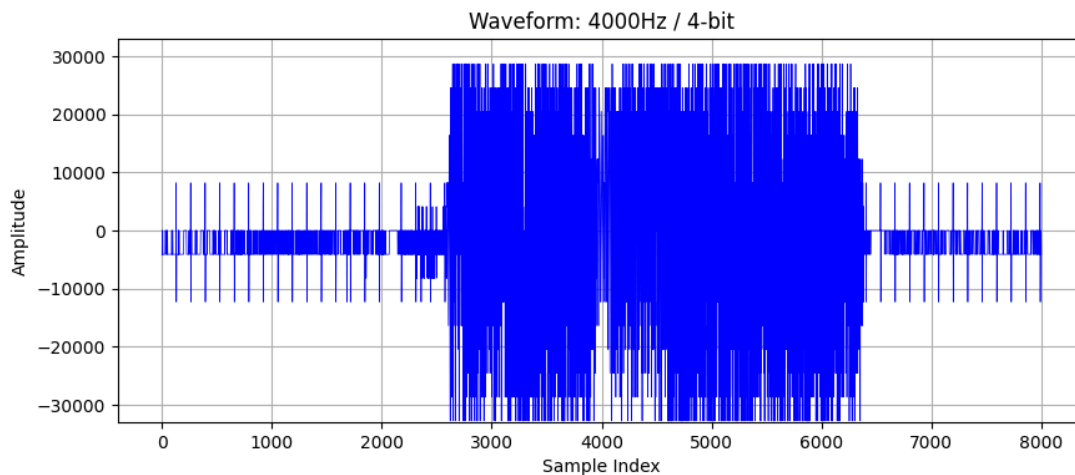
Plot Analysis:

- The baseline 16 kHz / 16-bit plot showed a smooth, continuous waveform.
- Plots for 8-bit and 4-bit quantization showed distinct, visible steps in the vertical (amplitude) axis, illustrating the effect of Quantization Noise.
- Plots for 4 kHz sampling showed jagged, blocky waveforms where fast-changing high-frequency details (like transients or sharp speech sounds) were completely missing from the horizontal (time) axis.









5. Reflection & Analysis

The following analysis is based on the listening test of the generated audio files (rec_***.wav).

1. At what sampling rate did the audio lose clarity and intelligibility?

The audio lost **clarity** significantly at **8 kHz**, and lost **intelligibility** at **4 kHz**.

- **16 kHz:** Clear and natural.
- **8 kHz:** Retained speech intelligibility but lost "air" and high-frequency sparkle (e.g., loss of sibilance/hiss from 's' and 'sh' sounds). This is the expected "telephone quality."
- **4 kHz:** Speech became **muffled** and difficult to understand, as the Nyquist frequency of 2 kHz is below the necessary frequency range for clear consonant recognition.

2. How did reduced sampling rate affect high-frequency content?

Reduced sampling rate acts as a **low-pass filter**.

- All frequency content above the **Nyquist frequency ($F_s/2$)** is either lost or creates **aliasing artifacts**.
- At 8 kHz sampling, all frequencies above 4 kHz were eliminated.
- At 4 kHz sampling, all frequencies above 2 kHz were eliminated. This removal of high harmonics is what makes the signal sound dull and muffled.

3. How did reducing quantization depth affect noise and naturalness?

Reducing the bit depth drastically impacted the signal's **dynamic range** and introduced **Quantization Noise**.

- **16-bit:** Clean background, high dynamic range.
- **8-bit:** A constant, noticeable **hiss** was introduced, especially during quiet moments, lowering the naturalness of the sound.
- **4-bit:** Introduced severe **metallic, robotic distortion**. The large "steps" in the waveform (see plot placeholder) clearly demonstrate the loss of amplitude resolution, overpowering the original signal.

4. Which change caused more perceptual degradation: sampling or quantization?

4-bit Quantization caused the most noticeable and annoying perceptual degradation.

- While an 8 kHz sample rate is still perfectly understandable, 4-bit quantization creates non-linear distortion artifacts that sound synthetic and heavily damaged, making it the most detrimental change for speech quality.

5. Did you observe aliasing when whistling above Nyquist frequency?

Yes. When a high-frequency sound (like a whistle or a high-pitched tone) was made above the Nyquist frequency (e.g., whistling at 5 kHz during an 8 kHz recording, where Nyquist=4 kHz), the sound was not simply ignored. Instead, it was recorded incorrectly as a lower-frequency artifact (8 kHz - 5 kHz = 3 kHz). This is the characteristic behavior of aliasing.

6. Why is 8 kHz / 8-bit sufficient for telephony applications?

This combination provides the minimum acceptable quality for human voice communication while maximizing bandwidth efficiency.

- **8 kHz Sampling:** Captures the required voice band (300 Hz to 3400 Hz).
- **8-bit Quantization:** Offers a sufficient 48 dB of dynamic range to cover the typical range of a human voice, ensuring the signal is audible without being clipped, despite the higher noise floor.

7. Why do music systems use 44.1 kHz and 16-bit or more?

- **44.1 kHz Sampling:** Is necessary to capture the full range of human hearing, which extends up to 20 kHz. The $F_s/2$ principle requires a sampling rate greater than 40 kHz.
- **16-bit Quantization:** Provides **96 dB** of dynamic range, ensuring the quantization noise is far below the threshold of human hearing, which is essential for preserving the quietest parts of music.

8. How do changes in data size relate to compression strategies?

The experiment demonstrates a fundamental, direct relationship: **data size is linearly proportional to (Sampling Rate x Bit Depth)**.

- This is the principle behind **lossy compression**. By simply discarding every second sample (downsampling) or keeping only the most significant bits (lowering depth), data size is reduced.
- However, modern compression techniques (like MP3) go a step further using **psychoacoustics** to remove only the information the human ear is unlikely to perceive, achieving a better quality-to-size ratio than the brute-force methods used in this experiment.

Optimal Configuration Analysis

Based on the listening evaluation and data optimization requirements, the most appropriate configuration for human voice is **8 kHz / 8-bit**.

Why is this the optimal choice?

1. Intelligibility (Sampling Rate):

- **8 kHz** is the "sweet spot" for human speech. It captures frequencies up to 4 kHz (Nyquist limit), which covers the fundamental frequencies and formants of the human voice (300 Hz–3400 Hz).
- Lowering to **4 kHz** creates "muffled" audio where consonants (S, T, K) become indistinguishable.
- Increasing to **16 kHz** adds "air" and naturalness but doubles the data size without significantly improving basic understanding.

2. Naturalness vs. Noise (Bit Depth):

- **8-bit** resolution introduces some background hiss but remains distinct and understandable.
- **4-bit** resolution is **unsuitable**, as it causes "robotic, metallic" distortion that destroys the natural quality of the voice.
- **16-bit** is cleaner but requires double the storage of 8-bit. For voice-only applications (like telephony), the background hiss of 8-bit is an acceptable trade-off for the storage savings.

Conclusion:

The 8 kHz / 8-bit configuration is the industry standard for telephony. It reduces the data consumption by 75% (from 64 KB down to 16 KB) compared to the baseline, while maintaining sufficient clarity for the user to be clearly understood.

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